A rotating condensate tracking $V \propto r^n$ cannot reach kination: exact equation of state $w = (n-2)/(n+2)$

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A complex scalar with a conserved charge Q, written in polar form as a rotating condensate of amplitude r, has rotational energy density that would scale as a^{-6} (kination, $w = 1$) if r were held fixed. On a polynomial potential $V \propto r^n$ the amplitude does not stay fixed: it tracks the minimum of the effective radial potential formed by V plus the centrifugal barrier. Tracking yields the exact equation of state

$$w = \frac{n-2}{n+2},$$

strictly below unity for every finite polynomial degree n, and equal to the kination limit only as $n \to \infty$. Freeze-out of r (the only escape to $w = 1$) requires a trans-Planckian amplitude, $r \gtrsim \mathcal{O}(1)\, M_{\rm Pl}$. Numerical integration of the radial field equation through a contracting background reproduces the analytic w to a few parts in 10^5 at $n = 2, 4, 6$. The result is classical field theory with a conserved charge; it is independent of any particular dark-sector model. It is a negative sector result: a rotating condensate of this class cannot supply the stiff phase often invoked to survive a BKL approach to a crunch.

I. MOTIVATION

In a contracting cosmology the shear energy density of anisotropic modes scales as a^{-6}. Radiation ($w = 1/3$) scales as a^{-4}, so the anisotropy-to-radiation ratio grows as a^{-2} and becomes catastrophic over even a few decades of contraction. The standard route past a Belinski–Khalatnikov–Lifshitz (BKL) approach [1, 2] is a stiff phase with $w \geq 1$ (kination), which scales as a^{-6} and keeps the relative anisotropy from growing.

A rotating complex scalar looks like a natural supplier of that phase. Written $\Psi = (r/\sqrt{2})\, e^{i\theta}$ with conserved charge

$$Q = a^3 r^2 \dot{\theta}\,, \tag{1}$$

elimination of $\dot{\theta}$ at fixed Q produces a centrifugal term in the effective radial potential,

$$V_{\rm eff}(r) = V(r) + \frac{Q^2}{2a^6 r^2}\,. \tag{2}$$

If the amplitude r were held fixed, the rotational energy density $\rho_{\rm rot} = Q^2/(2a^6 r^2)$ would fall exactly as a^{-6} — pure kination. The question is whether r holds fixed on approach. That is decidable from the field equation alone.

This note answers that question for polynomial potentials $V \propto r^n$. The answer is negative at every sub-Planckian amplitude. No claim is made about bounce mechanisms, dark energy, or any model beyond the classical charged scalar.

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II. TRACKING EQUATION OF STATE

At the minimum of (2),

$$V'(r) = \frac{Q^2}{a^6 r^3}\,. \tag{3}$$

For $V = c\, r^n$ this integrates immediately to

$$r \propto a^{-6/(n+2)}\,. \tag{4}$$

The rotational kinetic energy is then related to the potential by

$$\rho_{\rm rot} = \frac{Q^2}{2a^6 r^2} = \frac{n}{2}\, V\,, \tag{5}$$

so the total density and pressure at tracking are

$$\rho = V + \rho_{\rm rot} = V\big(1 + n/2\big) = V\,(n+2)/2\,, \tag{6}$$

$$p = \rho_{\rm rot} - V = V\big(n/2 - 1\big) = V\,(n-2)/2\,. \tag{7}$$

Hence

$$\boxed{w = \frac{p}{\rho} = \frac{n-2}{n+2}}. \tag{8}$$

Equivalently, from (4), $\rho \propto a^{-6n/(n+2)} = a^{-3(1+w)}$, which recovers the same w.

Special cases reproduce textbook regimes rather than inventing new ones:

- $n = 2$ (quadratic / mass era): $w = 0$ (cold dust / oscillating massive scalar).
- $n = 4$ (quartic): $w = 1/3$ (radiation-like).
- $n \to \infty$: $w \to 1$ (kination approached only at infinite degree).

For every finite polynomial n, $w < 1$. No polynomial potential reaches the stiff condition by tracking.

The algebraic relation (8) is not new in cosmology: it is the standard effective equation of state of a coherently oscillating scalar in a monomial potential [3] (the cold-dark-matter reading of a massive scalar is the $n = 2$ case). Kination itself is the $w = 1$ end-member of kinetic domination [4]. What is stated here is the consequence for a *rotating* condensate whose centrifugal barrier might have been hoped to supply kination without freezing r.

III. WHY THE FIELD TRACKS RATHER THAN FREEZES

Freeze-out of r would leave $\rho_{\rm rot} \propto a^{-6}$ and restore $w = 1$. Freeze-out requires Hubble friction to beat the restoring curvature of $V_{\rm eff}$ at the tracking minimum, i.e. $V''_{\rm eff} \lesssim H^2$.

At the minimum (3), for $V = c\, r^n$,

$$V''_{\rm eff} = n(n+2)\,\frac{V}{r^2}\,, \qquad H^2 = \frac{\rho}{3M_{\rm Pl}^2} = \frac{V(n+2)}{6M_{\rm Pl}^2}\,, \tag{9}$$

so

$$\frac{V''_{\rm eff}}{H^2} = 6n\left(\frac{M_{\rm Pl}}{r}\right)^2. \tag{10}$$

The ratio is $\mathcal{O}(10)\,(M_{\rm Pl}/r)^2$. Freezing needs $V''_{\rm eff}/H^2 \sim 1$, hence

$$r \gtrsim \sqrt{6n}\,M_{\rm Pl}\,. \tag{11}$$

Numerically this is $r \gtrsim 3.5\,M_{\rm Pl}$ at $n = 2$ and $r \gtrsim 4.9\,M_{\rm Pl}$ at $n = 4$. For any *sub-Planckian* amplitude the curvature dominates by orders of magnitude, the field tracks, and w is pinned at (8). Kination from freeze-out is available only at trans-Planckian amplitude.

IV. NUMERICAL CHECK

The analytic result can be confirmed by integrating the radial field equation through a contracting FLRW background and measuring w from the slope of $\ln\rho$ against $\ln a$. In e-fold time $N = \ln a$ with $u = {\rm d}r/{\rm d}N$,

$$r'' + \big(3 + H'/H\big)r' + \frac{1}{H^2}\Big[V'(r) - \frac{Q^2}{a^6 r^3}\Big] = 0\,, \tag{12}$$

with H^2 recomputed each step from the Friedmann constraint. Starting on the tracking solution and contracting over several decades of a, one finds (sub-Planckian window):

n	w analytic	w integrated
2	0	0.000064
4	1/3	0.333395
6	1/2	0.500067

Agreement is at the level of a few parts in 10^5. The same integrations cross the trans-Planckian freeze threshold when r grows as a falls; above that threshold the measured w turns to 1.000 (to four figures), confirming that freeze-out does produce kination and that the price is the amplitude (10) already prices.

V. WHAT THIS DOES AND DOES NOT CLAIM

Does claim. For a rotating complex scalar with conserved Q tracking $V_{\rm eff}$ on $V \propto r^n$, the equation of state is exactly (8). No finite polynomial n reaches kination. Freeze-out to $w = 1$ requires $r \sim \mathcal{O}(M_{\rm Pl})$.

Does not claim. That no stiff component can exist in a contracting universe; only that *this* sector, at sub-Planckian amplitude, cannot supply one. Other fields, non-polynomial potentials, non-canonical kinetics, or a bounce mechanism that introduces a stiff component of its own are outside the argument. The note does not solve the BKL problem and does not construct a bounce.

The result is therefore a negative sector statement of the kind useful in model-building bookkeeping: if a construction relies on a rotating condensate of this class to protect a contracting phase against BKL growth, that reliance fails at every sub-Planckian amplitude for every polynomial potential.

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Independent numerical integration of the radial equation was used only to cross-check the analytic tracking law; the law itself follows from (3)–(8) without numerics.

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